

RichnessSelection: math and numerical recipe for optical cluster selection bias

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The mean lensing surface density around optically selected clusters of observed richness λ^{ob} at observed redshift z^{ob} is the sum of a one-halo term (the cluster’s own miscentered NFW profile) and a two-halo term (correlated neighbouring halos along the line of sight):

$$\langle \Sigma(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \langle \Sigma^{\text{1h}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle + \langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle, \quad (1)$$

where $\langle \Sigma^{\text{1h}} \rangle$ is evaluated from the miscentered NFW of the target halo and $\langle \Sigma^{\text{prj}} \rangle$ is the projected surface density that **RichnessSelection** implements. The physical origin of the selection effect is that REDMAPPER counts richness through red-sequence member galaxies, and red galaxies are biased tracers of large-scale structure: their correlation with the surrounding density field means that, at fixed halo mass, a line of sight with more red galaxies gets assigned a larger λ^{ob} than one with fewer. The two-halo term at fixed λ^{ob} therefore does not carry the bare halo bias $b(M, z)$; Costanzi et al. [2026] show it must instead carry a *selection-affected* cluster bias $b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$ that interpolates between a small-scale limit (dominated by close red-galaxy neighbours, $b_{\text{sel}}^{\text{small}}$) and a large-scale limit (set by correlated LSS at the red-galaxy population bias, $b_{\text{sel}}^{\text{large}}$). Concretely, rather than writing the two-halo term with the halo-bias aggregate b_{halo} as in a standard lensing pipeline, we factor the scale-dependent selection effect into b_{sel} ,

$$\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \int d\theta \sin \theta \int dz \frac{dV}{dz d\Omega} \int dM n(M, z) [1 + b(M, z) b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) \xi_{\text{NL}}] \Sigma_{\text{mis}}, \quad (2)$$

(the full form with all arguments is eq. (12)). The package returns b_{sel} and the resulting $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$ at the $(\lambda^{\text{ob}}, z^{\text{ob}})$ of a DES Y3 cluster bin in ~ 30 ms per call.

This note has three parts. The **model** section gives the mathematical pipeline that assembles b_{sel} non-circularly from halo and projection ingredients, and the resulting two-halo profile $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$. The **numerical pitfalls** section documents the two subtle hazards we hit when putting that pipeline on a grid — the θ -exclusion step at the halo-halo exclusion boundary and the photo- z band structure of the z -integrand — and the GL-on-physics-split fixes that drive the measured error below 0.01% vs `scipy.integrate.quad`. The **API and glossary** section is a pointer table from each mathematical symbol to the Python call that returns it.

Model

1 Scope and notation

`RichnessSelection` computes the two-halo projected profile $\langle \Sigma^{\text{prj}}(R \mid \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$ (eq. 12) together with the selection bias b_{sel} that feeds it, at one $(\lambda^{\text{ob}}, z^{\text{ob}})$ per call. Photo- z on the target, target miscentering, triaxiality, and REDMAPPER percolation member-loss are left out (only the $\lambda < \lambda^{\text{ob}}$ ranking cut is kept). Linear $P(k, z)$ from CAMB feeds ξ_{NL} via haloFit + FFTlog.

Masses are in $h^{-1} M_{\odot}$; R is a physical projected distance; θ is a sky angle; χ , R_{λ} , R_{excl} are comoving in h^{-1} Mpc. Default cosmology is Buzzard-v1.1 ($\Omega_m = 0.286$, $h = 0.7$, $\sigma_8 = 0.82$). Key shorthand: $R_{\lambda}(\lambda^{\text{ob}}) = (\lambda^{\text{ob}}/100)^{0.2} h^{-1}$ Mpc, $R_{\text{excl}} \equiv R_{\lambda}(\lambda^{\text{ob}})(1 + z^{\text{ob}})$, and $b_{\text{sel}}^{\text{small}}/b_{\text{sel}}^{\text{large}}$ are the two asymptotes of the sigmoid ansatz below.

2 Core equations

2.1 Halo-bias aggregate $b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$

A single 1-D $\ln M$ integral at fixed z^{ob} , weighting the Tinker-2010 halo bias by the halo-mass distribution at fixed observed richness:

$$b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \frac{\int d \ln M \, M \, n(M, z^{\text{ob}}) \, b(M, z^{\text{ob}}) \, P(\lambda^{\text{ob}} \mid M, z^{\text{ob}})}{\int d \ln M \, M \, n(M, z^{\text{ob}}) \, P(\lambda^{\text{ob}} \mid M, z^{\text{ob}})}. \quad (3)$$

Here M is the halo mass in $h^{-1} M_{\odot}$; $\ln M$ runs over the integration support $[10^{13}, 10^{15.5}] h^{-1} M_{\odot}$; $n(M, z)$ is the Tinker-2008 halo mass function (§6); $b(M, z)$ is the Tinker-2010 peak-height halo bias; and $P(\lambda^{\text{ob}} \mid M, z)$ is the mass–observable relation pdf from `MOR`. The factor of M inside the integrand converts dM to $d \ln M$.

2.2 Sigmoid ansatz for $b_{\text{sel}}(\theta)$

Costanzi et al. [2026], eq. `b_sel_theta`:

$$b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta) = b_{\text{sel}}^{\text{small}} [1 - \sigma(\theta)] + b_{\text{sel}}^{\text{large}} \sigma(\theta), \quad \sigma(\theta) = \frac{1}{1 + e^{-k(\theta - \theta_0)}}. \quad (4)$$

Here θ is the sky angular separation from the target; $b_{\text{sel}}^{\text{small}}$ and $b_{\text{sel}}^{\text{large}}$ are the two plateaus (small- and large-angle limits of the bias, both scalars at fixed $(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$); $\sigma(\theta)$ is the logistic transition function whose shape constants are fixed to $k = 2.5/\theta_{\lambda}$ and $\theta_0 = \theta_{\lambda}/2$, with $\theta_{\lambda} \equiv R_{\lambda}(\lambda^{\text{ob}})/D_A(z^{\text{ob}})$ the angle subtended by the target’s REDMAPPER radius $R_{\lambda}(\lambda^{\text{ob}}) = (\lambda^{\text{ob}}/100)^{0.2} h^{-1}$ Mpc. The θ -shape is fixed analytically; only the two plateaus $b_{\text{sel}}^{\text{small}}$ and $b_{\text{sel}}^{\text{large}}$ need to be computed.

2.3 Projection kernel and $\mathcal{P}[X]$ operator

The *projection kernel* ρ^{prj} is the per-halo weight with which a projected halo at (λ, z, θ) contaminates the observed richness of a target at $(\lambda^{\text{ob}}, z^{\text{ob}})$ (Costanzi et al. 2026, eq. `weight`):

$$\rho^{\text{prj}}(\lambda, z, \theta \mid \lambda^{\text{ob}}, z^{\text{ob}}) = w_z(z, z^{\text{ob}}) f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}. \quad (5)$$

Here λ is the projected halo’s true richness; z is its redshift; θ is its angular separation from the target; $w_z(z, z^{\text{ob}})$ is the parabolic photo- z kernel quantifying the probability that a galaxy at redshift

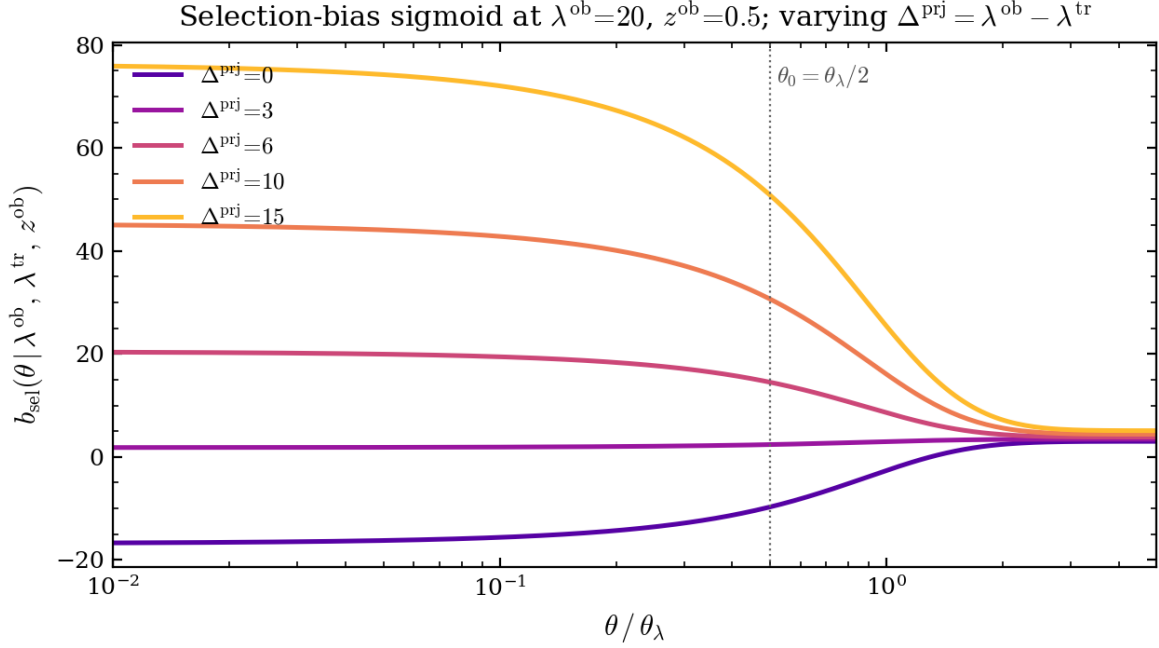


Figure 1: Selection-bias sigmoid $b_{\text{sel}}(\theta | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$ at fixed $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ as we vary the projection contamination $\Delta^{\text{prj}} \equiv \lambda^{\text{ob}} - \lambda^{\text{tr}}$. The x -axis is scaled by the per-bin angle θ_λ , so the sigmoid pivot $\theta_0 = \theta_\lambda/2$ (dotted) is universal. The θ -shape is fixed by the ansatz; Δ^{prj} drives both plateaus through Steps 4–5. The left plateau $b_{\text{sel}}^{\text{small}}$ (close neighbours projecting into the target aperture) grows fastest with Δ^{prj} ; the right plateau $b_{\text{sel}}^{\text{large}}$ (correlated large-scale structure) moves less because it is closed by the empirical $b_{\text{halo}}[1 + 0.13 \delta^{\text{prj}}]$ relation.

z is counted in the target’s aperture at z^{ob} ; $f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) \in [0, 1]$ is the closed-form overlap fraction of the two REDMAPPER disks (of the target and the projected halo); the factor λ converts “this halo is present” into “this halo contributes λ counts”; and the indicator $\mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}$ implements the REDMAPPER ranking cut (only projected halos of lower richness than the target contaminate the target’s aperture).

The $\mathcal{P}[X]$ operator is the line-of-sight-averaged integral of an arbitrary $X(M, z, \theta)$ weighted by the halo distribution and ρ^{prj} :

$$\begin{aligned} \mathcal{P}[X](\lambda^{\text{ob}}, z^{\text{ob}}) &\equiv \int dz \frac{dV}{dz d\Omega}(z) \int dM n(M, z) \int d\lambda P(\lambda | M, z) \\ &\cdot 2\pi \int d\theta \sin \theta \rho^{\text{prj}}(\lambda, z, \theta | \lambda^{\text{ob}}, z^{\text{ob}}) X(M, z, \theta | z^{\text{ob}}). \end{aligned} \quad (6)$$

The integrations marginalise, in order, over the projected halo’s redshift z (weighted by comoving volume per steradian $dV/dz d\Omega$), mass M (weighted by the halo mass function $n(M, z)$), true richness λ (weighted by the MOR $P(\lambda | M, z)$), and angular separation θ from the target (with the usual $2\pi \sin \theta d\theta$ sky-element).

Three specialisations drive the pipeline:

$$\mathcal{P}[1] = \langle \Delta_{\text{bkg}}^{\text{prj}} \rangle(\lambda^{\text{ob}}, z^{\text{ob}}), \quad I_2 \equiv \mathcal{P}[b \xi_{\text{NL}}], \quad I_1 \equiv \mathcal{P}[b \xi_{\text{NL}} \sigma(\theta)], \quad (7)$$

where $b = b(M, z)$ is the Tinker-2010 halo bias (the halo’s own clustering amplitude) and $\xi_{\text{NL}} = \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)$ is the nonlinear matter correlation function evaluated at the 3-D comoving separation $\Delta\chi(z, z^{\text{ob}}, \theta)$ between the target at z^{ob} and the projected halo at (z, θ) . By linearity, $\mathcal{P}[b \xi_{\text{NL}}(1 - \sigma)] = I_2 - I_1$. Halo exclusion sets $\xi_{\text{NL}} \equiv 0$ whenever $\Delta\chi < R_{\text{excl}}$.

2.4 Bias pipeline

This subsection gives the calculation chain from halo and projection ingredients to the selection-affected cluster bias $b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$ that eq. (12) requires. From the halo ingredients $\{n(M, z), b(M, z), P(\lambda | M, z)\}$ and the projection ingredients $\{w_z, f_A, \xi_{\text{NL}}\}$ it first evaluates the halo-bias aggregate $b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$ of eq. (3), a 1-D $\ln M$ integral at fixed z^{ob} . In parallel, the hardest numerical block computes the three forward integrals $\mathcal{P}[1]$, I_1 , I_2 of eq. (6); the recipes for this step are the subject of §3–4 below. With these in hand, the random-aperture projection contamination

$$\Delta_{\text{RND}}^{\text{prj}}(\lambda^{\text{ob}}, z^{\text{ob}}) = \mathcal{P}[1](\lambda^{\text{ob}}, z^{\text{ob}}) + b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) I_2(\lambda^{\text{ob}}, z^{\text{ob}}) \quad (8)$$

follows by substituting $b_{\text{sel}} \rightarrow b_{\text{halo}}$ in the b_{sel} -linearised identity (b_{halo} pulls out of the mass integral because it is mass-independent). The large- θ plateau is then

$$b_{\text{sel}}^{\text{large}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) = b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}}) [1 + 0.13 \delta^{\text{prj}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})], \quad \delta^{\text{prj}} \equiv \frac{\lambda^{\text{ob}} - \lambda^{\text{tr}}}{\Delta_{\text{RND}}^{\text{prj}}} - 1, \quad (9)$$

where δ^{prj} is the fractional excess contamination at $(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}})$ relative to a random aperture and the empirical coefficient 0.13 is the Buzzard calibration. Finally the small- θ plateau closes the remaining linear equation in b_{sel} :

$$b_{\text{sel}}^{\text{small}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) = \frac{(\lambda^{\text{ob}} - \lambda^{\text{tr}}) - \mathcal{P}[1] - b_{\text{sel}}^{\text{large}} I_1}{I_2 - I_1}, \quad (10)$$

where the denominator $I_2 - I_1 = \mathcal{P}[b \xi_{\text{NL}}(1 - \sigma)]$ is the contribution of the $1 - \sigma$ (small- θ) part of the sigmoid ansatz, and the numerator is what remains of the Δ^{prj} identity once the $\mathcal{P}[1]$ constant and the $b_{\text{sel}}^{\text{large}}$ -weighted I_1 piece have been subtracted. With $b_{\text{sel}}^{\text{small}}$ and $b_{\text{sel}}^{\text{large}}$ both known, eq. (4) returns $b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta)$. Marginalising over λ^{tr} via

$$b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) = \int d\lambda^{\text{tr}} b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta) P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}), \quad (11)$$

with $P(\lambda^{\text{tr}} | \lambda^{\text{ob}}, z^{\text{ob}}) \propto P(\lambda^{\text{ob}} | \lambda^{\text{tr}}, z^{\text{ob}}) \pi(\lambda^{\text{tr}}, z^{\text{ob}})$ and $\pi(\lambda^{\text{tr}}, z^{\text{ob}}) = \int dM n(M, z) P(\lambda^{\text{tr}} | M, z)$ the prior on the true richness, gives the final input to eq. (12).

In **SelBias** (§4), steps computed once per $(\lambda^{\text{ob}}, z^{\text{ob}})$ (b_{halo} , $\mathcal{P}[1]$, I_1 , I_2 , $\Delta_{\text{RND}}^{\text{prj}}$) are cached inside **bias_precompute**; the per- λ^{tr} closures of eqs. (9)–(10) live in **bias_from_precomp**; and the sigmoid assembly + λ^{tr} -marginalisation are **b_lob_theta** and **b_sel_marginalised** respectively.

2.5 Two-halo projected profile

Costanzi et al. [2026], eq. **sigprj**:

$$\langle \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, z^{\text{ob}}) \rangle = \int d\theta \sin \theta \int dz \frac{dV}{dz d\Omega} \int dM n(M, z) \cdot [1 + b(M, z) b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)] \Sigma_{\text{mis}}(R | M, z, \theta, z^{\text{ob}}). \quad (12)$$

Here R is the physical projected distance around the target in h^{-1} Mpc; θ is the angular separation of the contaminating halo from the target; z is the halo's redshift; M its mass; $n(M, z)$ the halo mass function; $b(M, z)$ the halo bias; b_{sel} the selection-affected cluster bias from eq. (11); ξ_{NL} the nonlinear matter correlation function at the 3-D separation between the target and the (z, θ) halo; and Σ_{mis} is the azimuth-averaged miscentered NFW surface density. The 1 inside the bracket is the mean

(uncorrelated) contribution; the $b b_{\text{sel}} \xi_{\text{NL}}$ term is the correlation-function boost that the two-halo term captures.

Σ_{mis} is the one-halo surface density integrated over the azimuthal angle φ because the projected halo is off-centre with respect to the profile radius:

$$\Sigma_{\text{mis}}(R | M, z, \theta, z^{\text{ob}}) = \int_0^{2\pi} d\varphi \Sigma(R_h | M, z), \quad R_h = \sqrt{R^2 + R_\theta^2 - 2RR_\theta \cos \varphi}, \quad (13)$$

with $R_\theta \equiv \theta D_A(z^{\text{ob}})$ the physical projected distance from the target to the contaminating halo and $\Sigma(R_h | M, z)$ the centred NFW surface density of the contaminating halo at 2-D radius R_h from its own centre. Σ_{mis} is tabulated from the offset-NFW lookup (`nfw.py`), with a factor of 2 to match the $\int_0^{2\pi}$ convention.

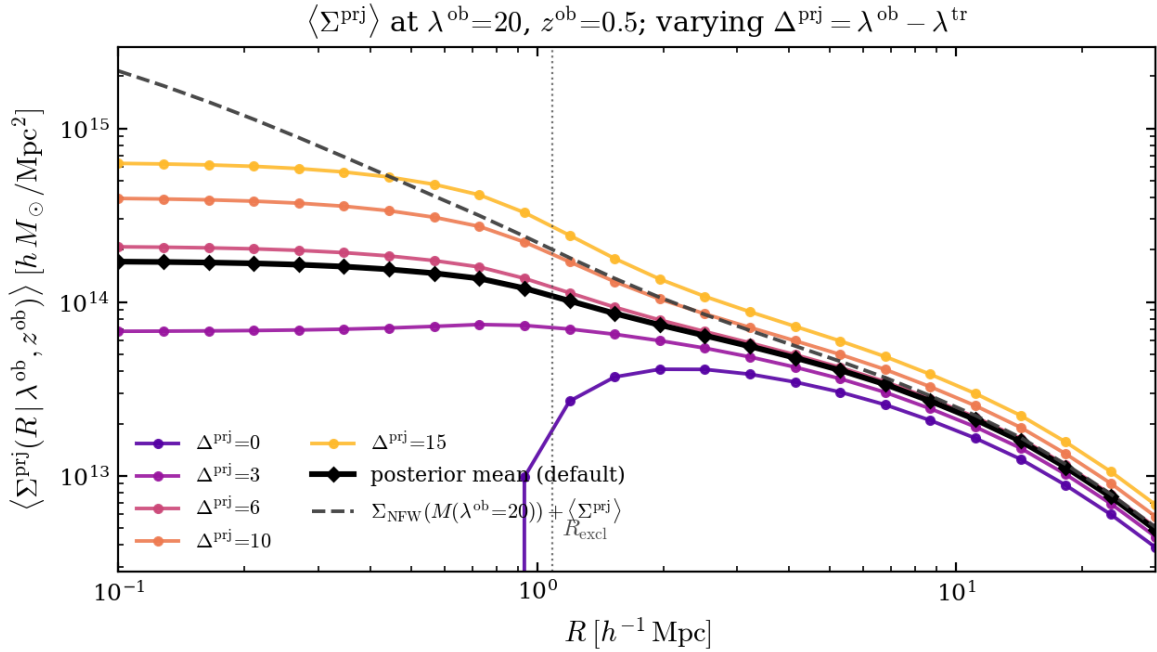


Figure 2: Two-halo $\langle \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$ at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, varying the projection contamination $\Delta^{\text{prj}} \equiv \lambda^{\text{ob}} - \lambda^{\text{tr}}$ (same colour palette as Fig. 1). b_{sel} is evaluated at fixed λ^{tr} (not λ^{tr} -marginalised) to match the Fig. 1 exercise. Dotted line: $R_{\text{excl}} = R_\lambda(\lambda^{\text{ob}})(1 + z^{\text{ob}})$, separating the miscentered-NFW-dominated inner regime from the correlated $b_{\text{sel}} \cdot \xi_{\text{NL}}$ large-scale regime. $\Delta^{\text{prj}} = 0$ sits anomalously low at small R because both b_{sel} plateaus collapse to the halo-bias aggregate in that limit.

2.6 Lensing excess $\langle \Delta \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$

The two-halo observable returned by `DeltaSigmaPrj` is the radial excess of (12),

$$\langle \Delta \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, z^{\text{ob}}) \rangle \equiv \bar{\Sigma}^{\text{prj}}(< R | \lambda^{\text{ob}}, z^{\text{ob}}) - \langle \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, z^{\text{ob}}) \rangle. \quad (14)$$

Because $R \rightarrow \bar{\Sigma}(< R) - \Sigma(R)$ is a linear functional in the radial argument only, it commutes with the (θ, z, M) integrals in eq. (12): the pipeline of §3–4 carries over unchanged, and the only substitution is at the NFW lookup, $\Sigma_{\text{mis}} \rightarrow \Delta \Sigma_{\text{mis}} \equiv \bar{\Sigma}_{\text{mis}}(< R) - \Sigma_{\text{mis}}(R)$, evaluated via the companion `log_deltasigma_single` table shipped with the offset-NFW kit. The full derivation, the scale argument for adaptive θ_{max} , and the factor-of-2 convention cross-check against the analytical Wright & Brainerd 2000 centred NFW $\Delta \Sigma$ live in `docs/delta_sigma_prj_derivation.tex`.

Profile vs projection contamination. Figure 3 shows $\langle \Delta \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$ at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, varying $\Delta^{\text{prj}} = \lambda^{\text{ob}} - \lambda^{\text{tr}}$ — the same sweep as Fig. 1 (b_{sel} -sigmoid) and Fig. 2 ($\langle \Sigma^{\text{prj}} \rangle$). The shape is the familiar two-halo lensing bump peaking around $\sim R_\lambda$ (dotted R_{excl} marker); amplitude scales with Δ^{prj} because both b_{sel} plateaus ($b_{\text{sel}}^{\text{small}}, b_{\text{sel}}^{\text{large}}$) grow with projection contamination through Steps 4–5 of the bias pipeline.

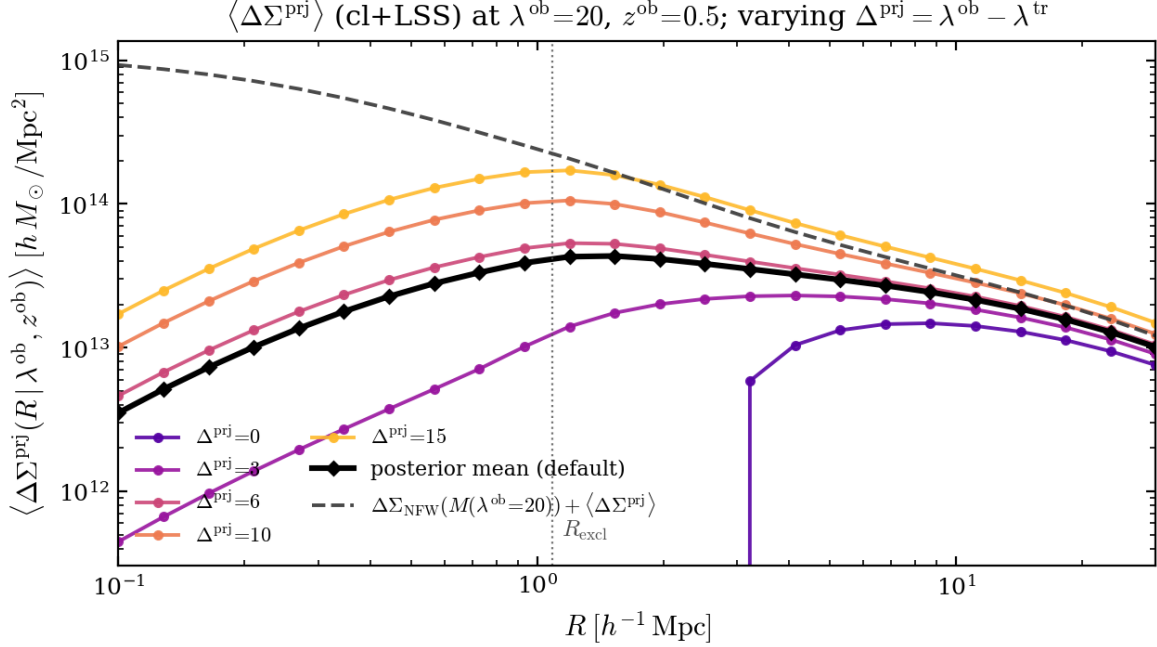


Figure 3: $\langle \Delta \Sigma^{\text{prj}}(R | \lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}) \rangle$ at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$, varying $\Delta^{\text{prj}} = \lambda^{\text{ob}} - \lambda^{\text{tr}}$ (c1+LSS correlation-excess piece, the class default). Dotted line: $R_{\text{excl}} = R_\lambda(\lambda^{\text{ob}})(1 + z^{\text{ob}})$. Inner cut-off from the LoS-slab exclusion; outer decay set by the joint support of $\Delta \Sigma_{\text{mis}}(R | R_{\text{mis}} \sim R)$ and $\xi_{\text{NL}}(r)$.

Precision vs scipy.quad. Figure 4 compares the production `DeltaSigmaPrj` ($n_{\text{per_seg}} = 30$, $N_M = 24$, $N_z = 80$, $R_{\text{max}} = 30 h^{-1} \text{Mpc}$) against a nested-`scipy.quad` reference built by `validations/delta_sigma_prj_diagnostics.py` at the same $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$ and $R \in \{0.3, 1, 3, 10\} h^{-1} \text{Mpc}$. The reference is parallelised across the 8 (R, kind) jobs via `multiprocessing.Pool` (8-core wall clock ~ 3 min vs ~ 35 min serial); quad tolerances $\epsilon_\theta = 3 \times 10^{-3}$, $\epsilon_z = 10^{-2}$. The residual panel shows sub-0.5% agreement on $R \geq 1 h^{-1} \text{Mpc}$ on both **total** and **c1**. At $R = 0.3$ the residual reaches $\sim 2\%$: the $[1 + b_{\text{sel}} \xi_{\text{NL}}]$ bracket changes amplitude sharply where $\theta_{\text{excl}}(z)$ and $\theta_R = R/D_A(z^{\text{ob}})$ coincide, and `scipy.quad`'s adaptive refinement issues an `IntegrationWarning` at that crossing. Tightening the quad tolerances by $3\times$ shifts the reference by $< 0.01\%$ (so the quad itself is converged), and the GL-on-segments production side is self-converged to $< 0.5\%$ between $n_{\text{per_seg}} = 30$ and 120: the residual at $R = 0.3$ is therefore a real quadrature-method difference at the smallest R , not a bug in either side. The regression test (`tests/test_delta_sigma_prj.py`) encodes this as a per- R tolerance (0.5% on $R \geq 1$, 2.5% at $R = 0.3$).

Numerical pitfalls

The three forward integrals that carry the whole pipeline — $\mathcal{P}[1]$, I_1 , I_2 of eq. (6) — are 4-D integrals over (z, θ, M, λ) . The inner (M, λ) axes are well-behaved and a plain Gauss–Legendre (GL) rule with $N_M = 24$ and $N_\lambda = 60$ converges them to $< 10^{-4}$; that part of the integration is boring. The outer

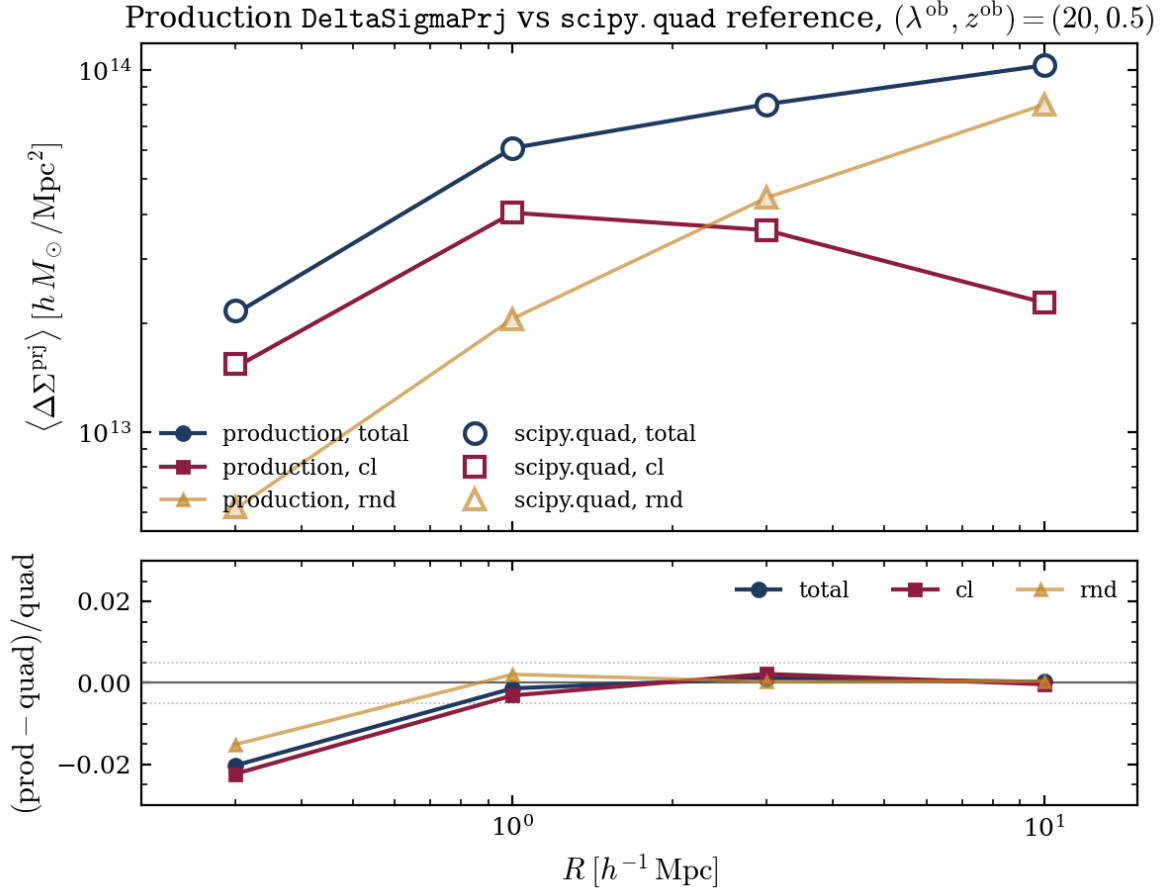


Figure 4: Precision audit of production `DeltaSigmaPrj` (filled markers, solid) against the `scipy.quad` reference (open markers) at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$. Top: absolute profiles for `total`, `cl` (default return), and `rnd` (truncation-dependent boundary term). Bottom: fractional residual $(\text{prod} - \text{quad}) / \text{quad}$, with $\pm 0.5\%$ guides (dotted). Agreement is sub-0.5% for $R \geq 1 h^{-1} \text{Mpc}$; at $R = 0.3$ the $\sim 2\%$ residual tracks a real quadrature-method difference where $\theta_{\text{excl}}(z) \sim \theta_R$ (see text).

(z, θ) axes, however, contain two sharp, physics-driven features that punish any uninformed fixed-rule method and were the source of the largest precision bugs we saw while developing this package:

- the **halo-exclusion step in θ** : at each z the integrand has a hard discontinuity at $\theta = \theta_{\text{excl}}(z)$, where ξ_{NL} is set to zero by the exclusion prescription. A fixed GL grid on $(0, 2\theta_{\lambda})$ with a mask integrates across this step and stalls at $\sim 0.3\%$ error regardless of N_{θ} (§3);
- the **exclusion-peak + ring-plateau structure in z** : the photo- z kernel carves out a window of width ~ 0.1 around z^{ob} , and inside that window ξ_{NL} produces twin sharp peaks at $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$ sitting on top of a narrow plateau — three qualitatively different regions crammed into $|z - z^{\text{ob}}| \lesssim 5 \times 10^{-4}$ (§4).

The recipes below move the GL nodes onto the physically correct coordinate in each region so that *the sharp features land at grid boundaries*, where GL rules naturally cluster nodes. The result is sub-0.01% precision on all three integrals at $N_z = 80$, $N_{\theta} = 10$ vs `scipy.integrate.quad` with matched inner grids. Total cost at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$: ~ 26 ms per `SelBias.bias_precompute`. If the bullets below already look familiar, the executive-summary version is: split the θ -axis at the exclusion boundary (per- z GL on $[\theta_{\text{excl}}(z), 2\theta_{\lambda}]$, no mask); split the z -axis by physics into a ring band + two outer halves, integrating in z on the ring and in $u = \ln |\Delta \chi_{\parallel}|$ on the outers.

3 The θ -axis: split at exclusion

What this section fixes. The θ -axis of $\mathcal{P}[1], I_1, I_2$ runs over the angular separation between the target and each projected halo. Halo exclusion makes this a discontinuous integrand, which breaks any fixed GL rule that sees the discontinuity as a noisy feature. Below we show why that happens and how moving the lower integration limit onto the discontinuity turns the problem into a smooth integral whose GL convergence is at its design rate.

At any z , halo exclusion kills θ below a threshold $\theta_{\text{excl}}(z)$ obtained from $\Delta\chi(z, \theta_{\text{excl}}) = R_{\text{excl}}$:

$$\cos \theta_{\text{excl}}(z) = \frac{\chi(z)^2 + \chi(z^{\text{ob}})^2 - R_{\text{excl}}^2}{2 \chi(z) \chi(z^{\text{ob}})}, \quad \text{clipped to } [-1, 1]. \quad (15)$$

A fixed-grid $N_\theta=20$ GL on $(0, 2\theta_\lambda)$ with $\xi_{\text{NL}} \rightarrow 0$ inside the step integrates across a discontinuity: the error oscillates in N_θ and settles to a $\sim 0.3\%$ floor. The fix is a per- z change of the integration interval: when $\theta_{\text{excl}}(z) > 0$, integrate on $[\theta_{\text{excl}}(z), 2\theta_\lambda]$ with no mask. GL nodes cluster at the endpoints where the step sits; the integrand is smooth on the interval.

N_θ	$\mathcal{P}[1]$ err	I_2 err	I_1 err
5	+0.060%	+0.046%	+0.046%
10	−0.005%	−0.001%	−0.004%
15	+0.000%	+0.000%	+0.000%

$N_\theta = 10$ reaches sub-0.01%: $\sim 12\times$ node reduction and two decades better precision than the mask recipe.

4 The z -axis: split by physics

What this section fixes. With θ handled, the outer z -integral becomes the dominant source of residual error — not because it is hard in absolute terms, but because its integrand has three different shapes over a $\sim 10^{-3}$ -wide window around z^{ob} . A single uniform z -grid under-samples the exclusion peaks; a single log-spaced grid in $|\Delta\chi|$ over-concentrates on the inner edge and misses the ring plateau. The fix is to integrate each physical region on the coordinate in which *that* region is smooth, and stitch them together.

Each of $\mathcal{P}[1], I_1, I_2$ factorises into an outer z -integral times an *inner* (M, λ, θ) -integral,

$$I = \int dz f_X(z), \quad f_X(z) \equiv \frac{dV}{dz d\Omega}(z) w_z(z, z^{\text{ob}}) \mathcal{I}_{\text{inner}}^X(z), \quad (16)$$

where the inner integral is the full (M, λ, θ) block of eq. (6) evaluated at a single z ,

$$\mathcal{I}_{\text{inner}}^X(z) \equiv \int dM n(M, z) \int d\lambda P(\lambda | M, z) \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}} \cdot 2\pi \int d\theta \sin \theta f_A(\theta, \lambda^{\text{ob}}, z^{\text{ob}}, \lambda, z) X(M, z, \theta | z^{\text{ob}}), \quad (17)$$

and X is the per-operator integrand label: $X = 1$ for $\mathcal{P}[1]$, $X = b(M, z) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta)$ for I_2 , and $X = b(M, z) \xi_{\text{NL}}(z, z^{\text{ob}}, \theta) \sigma(\theta)$ for I_1 . $f_X(z)$ is therefore the z -density that the outer integral has to quadrature.

For $\mathcal{P}[1]$, $X = 1$ contains no ξ_{NL} and no exclusion; $f_{\mathcal{P}[1]}(z)$ is a smooth single-peaked function of width $\sim \sigma_z(z^{\text{ob}})$ peaked at $z = z^{\text{ob}}$ (Fig. 5 left panel, dashed blue curve). For I_1, I_2 the halo-exclusion cut carves three regions. Defining $\delta z_{\text{excl}} \equiv R_{\text{excl}}/(c/H(z^{\text{ob}}))$ (at the reference point, $\approx 4.7 \times 10^{-4}$):

1. *Ring plateau* $|z - z^{\text{ob}}| < \delta z_{\text{excl}}$: the small- θ part is excluded; only $\theta > R_{\text{excl}}/\chi(z^{\text{ob}})$ contributes through a ring of structure at $\Delta\chi > R_{\text{excl}}$. Visible as a flat core of $f_{I_2}(z)$ between the dotted lines in Fig. 5 right panel.
2. *Exclusion peaks* at $|z - z^{\text{ob}}| \approx \delta z_{\text{excl}}$: ξ_{NL} is evaluated at $\Delta\chi \rightarrow R_{\text{excl}}$, its largest resolved value. $f_{I_2}(z)$ has two sharp peaks at $z = z^{\text{ob}} \pm \delta z_{\text{excl}}$, a few times the plateau amplitude (Fig. 5 right panel).
3. *Outer decay*: $f(z) \propto \Delta\chi_{\parallel}^{-1.7}$ modulated by w_z , truncating at $|z - z^{\text{ob}}| \sim \sigma_z(z^{\text{ob}}) \approx 0.09$ (Fig. 5 left panel, outer wings).

z -integrand at $\lambda^{\text{ob}}=20$, $z^{\text{ob}}=0.5$ ($\delta z_{\text{excl}} \approx 4.7e-04$)

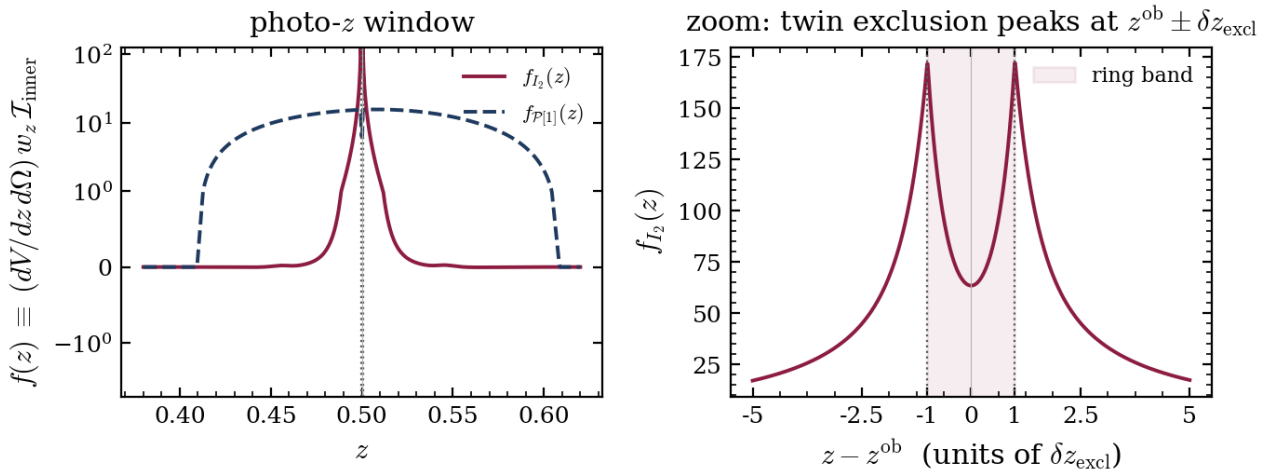


Figure 5: The z -integrand $f_X(z)$ of eq. (16) at $(\lambda^{\text{ob}}, z^{\text{ob}}) = (20, 0.5)$. *Left*: full photo- z window, symlog y -scale; the red solid curve is $f_{I_2}(z)$, the blue dashed curve is $f_{\mathcal{P}[1]}(z)$. $\mathcal{P}[1]$ (no ξ_{NL}) is smooth and single-peaked; I_2 has an additional two-peak exclusion substructure at $z^{\text{ob}} \pm \delta z_{\text{excl}}$ (dotted vertical lines) that is invisible at this scale but inflated by the symlog axis. *Right*: zoom to $|z - z^{\text{ob}}| < 5 \delta z_{\text{excl}}$, linear scale. The twin exclusion peaks are resolved; the ring-plateau band between them (red shading) is where ξ_{NL} is excluded and the integrand is flat. A naive uniform z -grid places zero nodes inside this window and under-samples both peaks, which is the hazard the split-by-physics recipe avoids.

Write $I = I_{\text{ring}} + I_{\text{outer}}^{\text{fg}} + I_{\text{outer}}^{\text{bg}}$ and integrate each region on a grid tailored to its shape. The ring band $(z^{\text{ob}} - \delta z_{\text{excl}}, z^{\text{ob}} + \delta z_{\text{excl}})$ uses GL in z (smooth integrand, $N_{\text{ring}} = \max(9, N_z/4)$). The outer pieces use GL in $u = \ln|\Delta\chi_{\parallel}|$, so $\xi_{\text{NL}} \propto e^{-1.7u}$ peaks at the inner endpoint $u = \ln R_{\text{excl}}$ where GL naturally clusters nodes. Per side: $N_{\text{outer}} = \max(15, (N_z - N_{\text{ring}})/2)$. Each region's integrand is smooth in its own variable; the sharp features sit at region boundaries where GL rules converge at their design rate.

At $N_z = 80$ this reaches sub-0.01% on $\mathcal{P}[1], I_1, I_2$ vs `scipy.integrate.quad` with matched inner grids (regression test `tests/test_integrals.py::TestSelBiasThetaSplit`). Edge cases (R_{excl} exceeds the photo- z support, or the ring band is wider than it) degrade gracefully via empty-array guards.

API and glossary

This section is a pointer table from the math above to the code that computes each quantity. The glossary table lists every symbol with its defining equation, a one-line description, and the Python call that returns it.

5 Objects

Cosmology (`cosmology.py`). Flat- Λ CDM background. Provides `chi(z) =  $\chi(z)$` comoving distance, `dV_dzdOm(z) =  $dV/dz d\Omega$` , and `h, H0`. Default: Buzzard-v1.1.

PkGrid (`pk.py`). CAMB-backed linear $P(k, z)$. One CAMB call per cosmology; re-evaluated in place for each (k, z) request.

SigmaM (`sigma_m.py`). $\sigma(M, z)$ top-hat variance via a Simpson rule on PkGrid’s k -grid. Used by both HMF and Bias.

HMF (`hmf.py`). Tinker 2008, $\Delta = 200$. `hmf(M, z)` returns $n(M, z)$ in $h^{-1} M_{\odot}^{-1} h^{-1} \text{Mpc}^{-3}$.

Bias (`bias.py`). Tinker 2010 peak-height halo bias. `bias(M, z)` returns $b(M, z)$.

MOR (`mor.py`). Mass–observable relation. `mor.pdf(lam, M, z)` returns $P(\lambda | M, z)$, the EMG kernel of Costanzi 2019.

XiNL (`xi_nl.py`). Nonlinear matter correlation function $\xi_{\text{NL}}(r, z)$ from halofit P_{NL} via an FFTlog Hankel transform. Lazy build (first call pays ~ 2 s, cached thereafter).

SelBias (`sel_bias.py`). The bias pipeline (§2.4). Holds references to all of the above. See the glossary below for per-method meanings.

NFWMiscentered (`nfw.py`). $\Sigma_{\text{mis}}(R | M, z, \theta, z^{\text{ob}})$ from the three offset-NFW lookup tables (Costanzi-notebook convention).

SigmaPrj (`sigma_prj.py`). The $\langle \Sigma^{\text{prj}} \rangle$ orchestrator (eq. 12). `SigmaPrj(cosmo, sel_bias, nfw)(R, lob, zob)` returns $\langle \Sigma^{\text{prj}}(R) \rangle$.

GridConfig (`config.py`). Grid-size defaults: $N_z = 80$, $N_{\theta} = 10$, $N_M = 24$. N_{λ} is a separate **SelBias** constructor arg (default 60).

6 Glossary of quantities

Rows group by role: halo ingredients, projection ingredients, the $\mathcal{P}$ operator specialisations, the bias-pipeline outputs, and the profile. “Eq.” links to the defining equation; “Code” gives the Python call.

Symbol	Meaning	Eq.	Code
<i>Halo ingredients</i>			
$n(M, z)$	Tinker-2008 halo mass function ($\Delta = 200$)	(3)	<code>HMF(sigma_m)(M, z)</code>
$b(M, z)$	Tinker-2010 peak-height halo bias	(3)	<code>Bias(sigma_m)(M, z)</code>
$P(\lambda M, z)$	MOR kernel (Costanzi-2019 EMG)	(6)	<code>MOR().pdf(lam, M, z)</code>
$\xi_{\text{NL}}(r, z)$	Nonlinear matter 2pt correlation (halofit+FFTlog); zero below R_{excl}	(6)	<code>XiNL(cosmo)(r, z)</code>
<i>Projection kernel and $\mathcal{P}[X]$ operator</i>			
ρ^{prj}	Per-halo projection weight $w_z f_A \lambda \mathbf{1}_{\{\lambda < \lambda^{\text{ob}}\}}$	(5)	internal to <code>SelBias.P_operator</code>
$\mathcal{P}[X]$	LoS average of X weighted by ρ^{prj} and halo distribution	(6)	<code>SelBias.P_operator(lob, zob)</code>
$\mathcal{P}[1] \equiv \langle \Delta_{\text{bkg}}^{\text{prj}} \rangle$	Mean background projection (no ξ_{NL} , smooth in z)	(6)	<code>bias_precompute(lob, zob) ['P1']</code>
$I_2 \equiv \mathcal{P}[b \xi_{\text{NL}}]$	Halo-bias-weighted ξ_{NL} ; drives $\Delta_{\text{RND}}^{\text{prj}}$	(6)	<code>bias_precompute(lob, zob) ['I2']</code>
$I_1 \equiv \mathcal{P}[b \xi_{\text{NL}} \sigma(\theta)]$	I_2 weighted by the sigmoid; enters Step 5	(6)	<code>bias_precompute(lob, zob) ['I1']</code>
<i>Bias-pipeline outputs</i>			
$b_{\text{halo}}(\lambda^{\text{ob}}, z^{\text{ob}})$	Halo-bias aggregate at fixed λ^{ob} (paper's b_{eff})	(3)	<code>SelBias.b_eff(lob, zob)</code>
$\Delta_{\text{RND}}^{\text{prj}}$	Projection contamination of a random aperture	(8)	<code>bias_precompute(lob, zob) ['Delta_RND']</code>
$b_{\text{sel}}^{\text{large}}$	Large- θ plateau (empirical closure)	(9)	<code>bias_from_precomp(pre, ltr) ['b_infty']</code>
$b_{\text{sel}}^{\text{small}}$	Small- θ plateau (Step 5 inversion)	(10)	<code>bias_from_precomp(pre, ltr) ['b_zero']</code>
$b_{\text{sel}}(\lambda^{\text{ob}}, \lambda^{\text{tr}}, z^{\text{ob}}, \theta)$	Scale-dependent bias via sigmoid ansatz	(4)	<code>SelBias.b_lob_theta(th, ltr, zob, lob, precomp)</code>
$b_{\text{sel}}(\lambda^{\text{ob}}, z^{\text{ob}}, \theta)$	λ^{tr} -marginalised bias; input to $\langle \Sigma^{\text{prj}} \rangle$	(11)	<code>SelBias.b_sel_marginalised(th, lob, zob, precomp)</code>
<i>Two-halo profile</i>			
$\Sigma_{\text{mis}}(R M, z, \theta, z^{\text{ob}})$	Azimuth-averaged miscentered NFW (offset-NFW lookup)	(13)	<code>NFWMiscentered(cosmo).sigma_grid(R, Rtheta)</code>
$\Delta \Sigma_{\text{mis}}$	Miscentered NFW radial excess $\bar{\Sigma}_{\text{mis}}(< R) - \Sigma_{\text{mis}}(R)$ (companion lookup table)	—	<code>NFWMiscentered.delta_sigma_grid</code>
$\langle \Sigma^{\text{prj}}(R \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$	Two-halo projected surface density (main output)	(12)	<code>SigmaPrj(cosmo, sel_bias, nfw)(R, lob, zob)</code>
$\langle \Delta \Sigma^{\text{prj}}(R \lambda^{\text{ob}}, z^{\text{ob}}) \rangle$	Two-halo lensing excess (see docs/delta_sigma_prj_derivation.tex)	—	<code>DeltaSigmaPrj(cosmo, sel_bias, nfw)(R, lob, zob)</code>

7 Validation

Unit tests. `tests/test_integrals.py`: `TestSelBiasThetaSplit` (sub-0.1% vs quad at $N_\theta = 10$), `TestLambdaAxisConverged` ($n_\lambda^{\text{tr}} = 30$ matches 100 to $< 10^{-4}$), `TestSigmaPrjTheta` (split-at- θ_R + monotonicity). `test_smoke.py`: end-to-end smoke test.

Validation scripts. `validations/`: `theta_convergence.py` sweeps N_θ ; `quad_matched_inner.py` compares the `_P_operator` output to `scipy.integrate.quad` with matched inner GL grids; `quad_validate.py` is the end-to-end quad cross-check.

Defaults. $N_z = 80$, $N_\theta = 10$, $N_M = 24$, $N_\lambda = 60$. No retuning is needed across the Y3 range ($\lambda^{\text{ob}} \in [20, 500]$, $z^{\text{ob}} \in [0.1, 0.8]$): only R_{excl} and $c/H(z^{\text{ob}})$ shift, and both are tracked analytically by the grid builder.

References

Costanzi, M., Wu, H.-Y., Esteves, J. H., Grandis, S., To, C., & Aguena, M. 2026, Phys. Rev. D (accepted), arXiv:2604.05833.